

Question ID 4e375d1f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 4e375d1f

SAT6 M 2.1

How many meters are equivalent to **2,300** centimeters? (**100 centimeters = 1 meter**)

- A. **0.043**
- B. **23**
- C. **2,400**
- D. **230,000**

ID: 4e375d1f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that 100 centimeters is equal to 1 meter. Therefore, 2,300 centimeters is equivalent to $2,300 \text{ centimeters} \times \frac{1 \text{ meter}}{100 \text{ centimeters}}$, or 23 meters.

Choice A is incorrect. 0.043 meters is equivalent to 4.3, not 2,300, centimeters.

Choice C is incorrect. 2,400 meters is equivalent to 240,000, not 2,300, centimeters.

Choice D is incorrect. 230,000 meters is equivalent to 23,000,000, not 2,300, centimeters.

Question Difficulty: Easy

Question ID bd9eb2b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bd9eb2b5

SAT6 M 2.2

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

- A. 8
- B. 9
- C. 64
- D. 80

ID: bd9eb2b5 Answer

Correct Answer: B

Rationale

Choice B is correct. Substituting 9 for x in the given function yields $72 = 8x$. Dividing each side of this equation by 8 yields $9 = x$. Therefore, $f(x) = 72$ when the value of x is 9.

Choice A is incorrect. This is the value of x for which $f(x) = 64$, not $f(x) = 72$.

Choice C is incorrect. This is the value of x for which $f(x) = 512$, not $f(x) = 72$.

Choice D is incorrect. This is the value of x for which $f(x) = 640$, not $f(x) = 72$.

Question Difficulty: Easy

Question ID 8784bc84

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 8784bc84

SAT6 M 2.3

What is 20% of 440?

- A. 44
- B. 88
- C. 880
- D. 1,760

ID: 8784bc84 Answer

Correct Answer: B

Rationale

Choice B is correct. 20% of 440 can be calculated as $\frac{20}{100}440$, which is equivalent to $\frac{8,800}{100}$, or 88.

Choice A is incorrect. This is 10%, not 20%, of 440. Choice C is incorrect. This is 200%, not 20%, of 440.

Choice D is incorrect. This is 400%, not 20%, of 440.

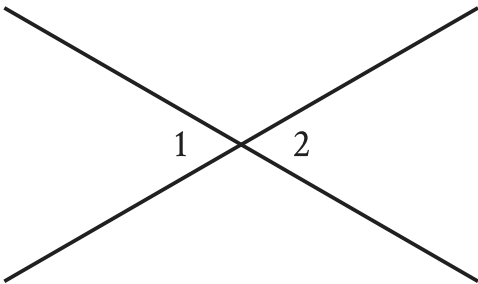
Question Difficulty: Easy

Question ID a456f28c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a456f28c

SAT6 M 2.4



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. Angle **1** and angle **2** are vertical angles. The measure of angle **1** is **72°**. What is the measure of angle **2**?

- A. **72°**
- B. 108°
- C. 144°
- D. 288°

ID: a456f28c Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that angle 1 and angle 2 are vertical angles, and the measure of angle 1 is 72°. Vertical angles have equal measures. Therefore, the measure of angle 2 is 72°.

Choice B is incorrect. This is the measure of an angle that is supplementary, not congruent, to angle 1.

Choice C is incorrect. This is the sum of the measures of angle 1 and angle 2.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 51aabd93

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 51aabd93

SAT6 M 2.5

$(p + 3) + 8 = 10$ What value of p is the solution to the given equation?

- A. -1
- B. 5
- C. 15
- D. 21

ID: 51aabd93 Answer

Correct Answer: A

Rationale

Choice A is correct. Subtracting 8 from both sides of the given equation yields $p + 3 = 2$. Subtracting 3 from both sides of this equation yields $p = -1$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

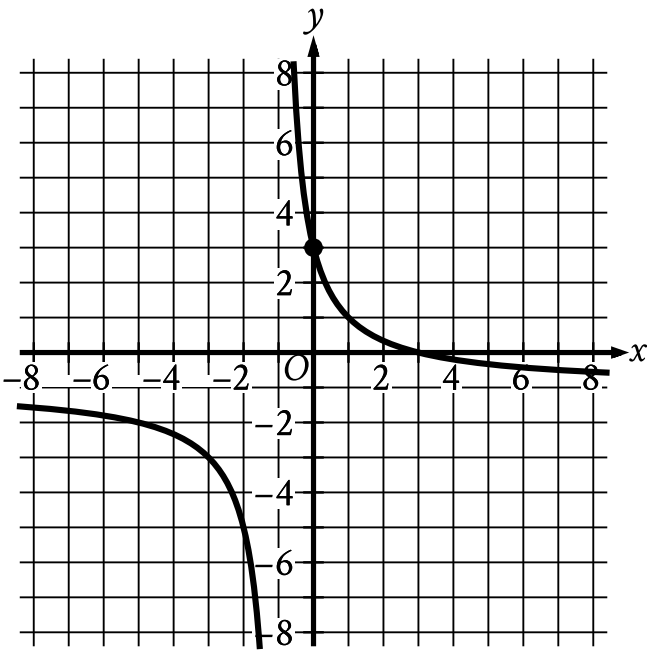
Question Difficulty: Easy

Question ID c99d154a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c99d154a

SAT6 M 2.6



What is the y -coordinate of the y -intercept of the graph shown?

ID: c99d154a Answer

Correct Answer: 3

Rationale

The correct answer is 3. A y -intercept of a graph in the xy -plane is a point x, y on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is 3. Therefore, the y -coordinate of the y -intercept of the graph shown is 3.

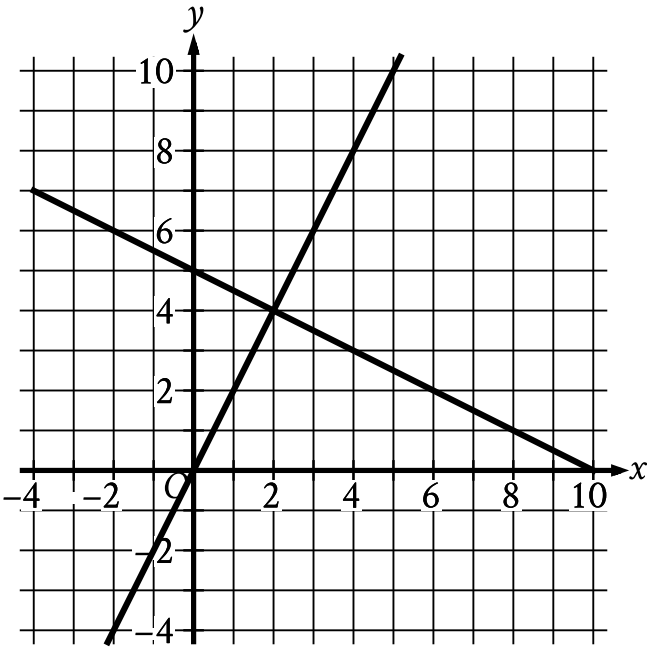
Question Difficulty: Easy

Question ID 1006cad7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 1006cad7

SAT6 M 2.7



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A. $(0, 5)$
- B. $(2, 4)$
- C. $(5, 10)$
- D. $(10, 0)$

ID: 1006cad7 Answer

Correct Answer: B

Rationale

Choice B is correct. A solution to a system of equations must be the solution to each equation in the system. It follows that if x, y is a solution to the system, the point x, y lies on the graph in the xy -plane of each equation in the system. The point that lies on each graph of the system of linear equations shown is their intersection point 2, 4. Therefore, the solution to the system is 2, 4.

Choice A is incorrect. The point 0, 5 lies on one, but not both, of the graphs of the linear equations shown.

Choice C is incorrect. The point 5, 10 lies on one, but not both, of the graphs of the linear equations shown.

Choice D is incorrect. The point 10, 0 lies on one, but not both, of the graphs of the linear equations shown.

Question Difficulty: Easy

Question ID 7399c3b0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 7399c3b0

SAT6 M 2.8

$k^2 - 53 = 91$ What is the positive solution to the given equation?

- A. 144
- B. 72
- C. 38
- D. 12

ID: 7399c3b0 Answer

Correct Answer: D

Rationale

Choice D is correct. Adding 53 to each side of the given equation yields $k^2 = 144$. Taking the square root of each side of this equation yields $k = \pm 12$. Therefore, the positive solution to the given equation is 12.

Choice A is incorrect. This is the positive solution to the equation $k^2 - 53 = 20,683$, not $k^2 - 53 = 91$.

Choice B is incorrect. This is the positive solution to the equation $k^2 - 53 = 5,131$, not $k^2 - 53 = 91$.

Choice C is incorrect. This is the positive solution to the equation $k^2 - 53 = 1,391$, not $k^2 - 53 = 91$.

Question Difficulty: Easy

Question ID aacc834b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: aacc834b

SAT6 M 2.9

$x = 8$ $x + 3y = 26$ The solution to the given system of equations is (x, y) . What is the value of y ?

ID: aacc834b Answer

Correct Answer: 6

Rationale

The correct answer is 6. The first equation in the given system is $x = 8$. Substituting 8 for x in the second equation in the given system yields $8 + 3y = 26$. Subtracting 8 from both sides of this equation yields $3y = 18$. Dividing both sides of this equation by 3 yields $y = 6$. Therefore, the value of y is 6.

Question Difficulty: Easy

Question ID 842cec4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 842cec4d

SAT6 M 2.10

During a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

- A. $s \leq 20$
- B. $s \leq 150$
- C. $s \leq 170$
- D. $150 \leq s \leq 170$

ID: 842cec4d Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that during a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. It's also given that s represents the cruising speed, in miles per hour, during this portion of the flight. It follows that the airplane's cruising speed, in miles per hour, was at least 150, which means $s \geq 150$, and was at most 170, which means $s \leq 170$. Therefore, the inequality that best represents this situation is $150 \leq s \leq 170$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

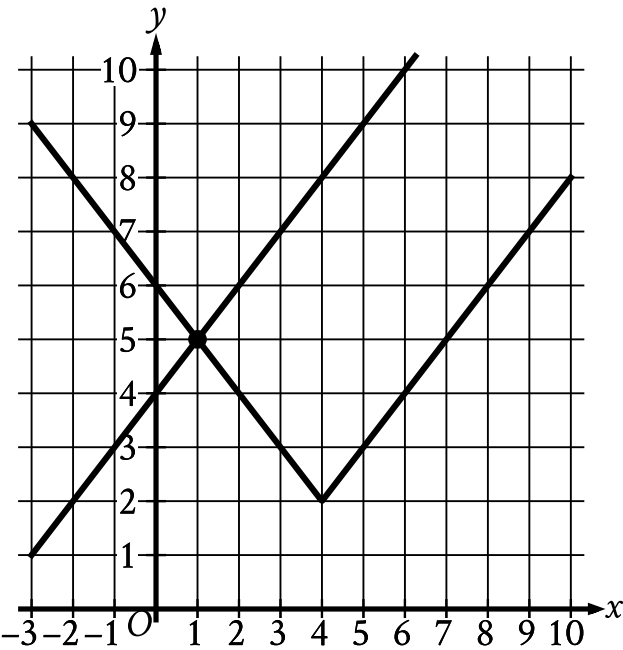
Question Difficulty: Easy

Question ID dd3a910a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: dd3a910a

SAT6 M 2.11



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

- A. $(-1, 5)$
- B. $(0, 4)$
- C. $(1, 5)$
- D. $(4, 2)$

ID: dd3a910a Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at the point 1, 5. Thus, the solution to the system is 1, 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the y -intercept of the graph of the linear function.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

Question Difficulty: Easy

Question ID c1eead73

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c1eead73

SAT6 M 2.12

The function g is defined by $g(x) = |x + 18|$. What is the value of $g(4)$?

- A. -18
- B. -4
- C. 14
- D. 22

ID: c1eead73 Answer

Correct Answer: D

Rationale

Choice D is correct. The value of $g4$ is the value of gx when $x = 4$. Substituting 4 for x in the given equation yields $g4 = 4 + 18$, which is equivalent to $g4 = 22$, or $g4 = 22$. Therefore, the value of $g4$ is 22.

Choice A is incorrect. This would be the value of $g4$ if function g was defined by $gx = -18$, not $gx = x + 18$.

Choice B is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x$, not $gx = x + 18$.

Choice C is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x + 18$, not $gx = x + 18$.

Question Difficulty: Easy

Question ID 9fc31513

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 9fc31513

SAT6 M 2.13

The perimeter of an isosceles triangle is **36** feet. Each of the two congruent sides of the triangle has a length of **10** feet. What is the length, in feet, of the third side?

- A. 10
- B. 12
- C. 16
- D. 18

ID: 9fc31513 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the perimeter of an isosceles triangle is 36 feet and that each of the two congruent sides has a length of 10 feet. The perimeter of a triangle is the sum of the lengths of its three sides. The equation $10 + 10 + x = 36$ can be used to represent this situation, where x is the length, in feet, of the third side. Combining like terms on the left-hand side of this equation yields $20 + x = 36$. Subtracting 20 from each side of this equation yields $x = 16$. Therefore, the length, in feet, of the third side is 16.

Choice A is incorrect. This would be the length, in feet, of the third side if the perimeter was 30 feet, not 36 feet.

Choice B is incorrect. This would be the length, in feet, of the third side if the perimeter was 32 feet, not 36 feet.

Choice D is incorrect. This would be the length, in feet, of the third side if the perimeter was 38 feet, not 36 feet.

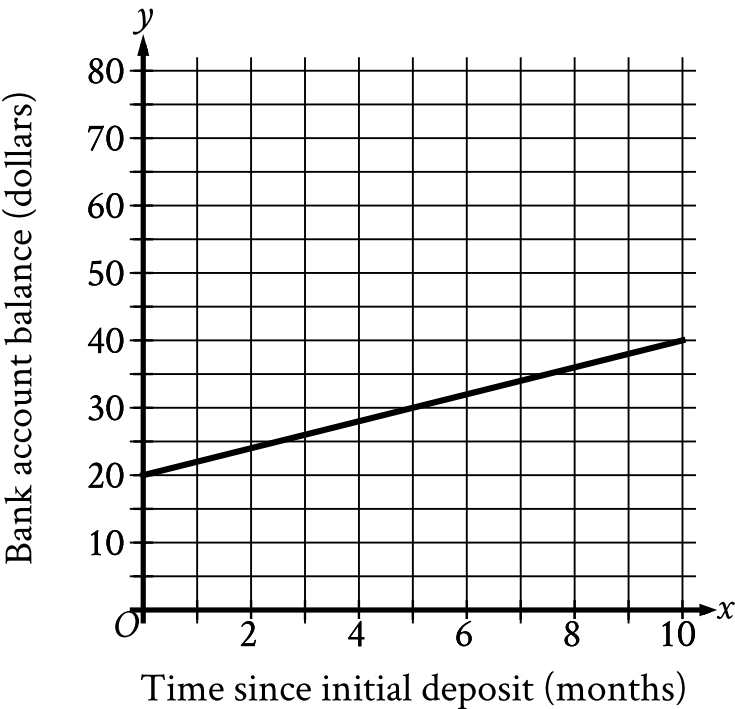
Question Difficulty: Easy

Question ID 2ecce641

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2ecce641

SAT6 M 2.14



A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

ID: 2ecce641 Answer

Correct Answer: 20

Rationale

The correct answer is 20. For the graph shown, the x -axis represents the time since the initial deposit, in months, and the y -axis represents the bank account balance, in dollars. The amount of the initial deposit is estimated by the y -coordinate of the point on the graph that represents 0 months since the initial deposit. Therefore, the amount of the initial deposit is estimated by the corresponding y -value for the point when $x = 0$. When $x = 0$, it is estimated that $y = 20$. Thus, the amount of the initial deposit estimated by the graph, to the nearest whole dollar, is 20.

Question Difficulty: Easy

Question ID 1fc4f47b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 1fc4f47b

SAT6 M 2.15

At a movie theater, there are a total of **350** customers. Each customer is located in either theater A, theater B, or theater C. If one of these customers is selected at random, the probability of selecting a customer who is located in theater A is **0.48**, and the probability of selecting a customer who is located in theater B is **0.24**. How many customers are located in theater C?

- A. 28
- B. 40
- C. 84
- D. 98

ID: 1fc4f47b Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that at a movie theater, there are a total of 350 customers and that each customer is located in either theater A, theater B, or theater C. If the probability of selecting a customer in theater A is 0.48, then 0.48×350 , or 168, customers are located in theater A. If the probability of selecting a customer in theater B is 0.24, then 0.24×350 , or 84, customers are located in theater B. It follows that there are $168 + 84$, or 252, customers in theater A and theater B. Therefore, there are $350 - 252$, or 98, customers in theater C.

Choice A is incorrect. This is the percent, not the number, of the customers that are located in theater C.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of customers that are located in theater B, not theater C.

Question Difficulty: Medium

Question ID ebd457f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: ebd457f4

SAT6 M 2.16

Brian saves $\frac{2}{5}$ of the \$215 he earns each week from his job. If Brian continues to save at this rate, how much money, in dollars, will Brian save in 9 weeks?

ID: ebd457f4 Answer

Correct Answer: 774

Rationale

The correct answer is 774. It's given that Brian saves $\frac{2}{5}$ of the \$ 215 he earns each week from his job. Therefore, Brian saves $\frac{2}{5}$ \$ 215, or \$ 86, per week. If Brian continues to save at this rate of \$ 86 per week for 9 weeks, then he will save a total of 986, or 774, dollars.

Question Difficulty: Medium

Question ID f28944ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f28944ff

SAT6 M 2.17

$q(x) = 32(2^x)$ Which table gives three values of x and their corresponding values of $q(x)$ for function q ?

- A.

x	-1	0	1
$q(x)$	-64	0	64
- B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64
- C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64
- D.

x	-1	0	1
$q(x)$	16	32	64

ID: f28944ff Answer

Correct Answer: D

Rationale

Choice D is correct. Substituting -1 for x in the given function yields $q(-1) = 32(2^{-1})$, which is equivalent to $q(-1) = 32\frac{1}{2}$, or $q(-1) = 16$. Therefore, when $x = -1$, the corresponding value of $q(x)$ for function q is 16. Substituting 0 for x in the given function yields $q(0) = 32(2^0)$, which is equivalent to $q(0) = 32(1)$, or $q(0) = 32$. Therefore, when $x = 0$, the corresponding value of $q(x)$ for function q is 32. Substituting 1 for x in the given function yields $q(1) = 32(2^1)$, which is equivalent to $q(1) = 32(2)$, or $q(1) = 64$. Therefore, when $x = 1$, the corresponding value of $q(x)$ for function q is 64. Of the choices given, only the table in choice D gives these three values of x and their corresponding values of $q(x)$ for function q .

Choice A is incorrect. This table gives three values of x and their corresponding values of $q(x)$ for the function $q(x) = 32x$.

Choice B is incorrect. This table gives three values of x and their corresponding values of $q(x)$ for the function $q(x) = 232^x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID ae33e0a1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: ae33e0a1

SAT6 M 2.18

A right circular cylinder has a base diameter of **22** centimeters and a height of **6** centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. **132** π
- B. **264** π
- C. **726** π
- D. **2,904** π

ID: ae33e0a1 Answer

Correct Answer: C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of 22 centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or 11, centimeters. Therefore, $r = 11$. Substituting 11 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi 11^2 6$, which is equivalent to $V = \pi 121 6$, or $V = 726 \pi$. Therefore, the volume, in cubic centimeters, of the cylinder is 726π .

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not 22, centimeters and a height of 6 centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not 22, centimeters and a height of 6 centimeters.

Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of 44, not 22, centimeters and a height of 6 centimeters.

Question Difficulty: Medium

Question ID e0f59119

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: e0f59119

SAT6 M 2.19

What is the slope of the graph of $y = \frac{1}{3}(29x + 10) + 5x$ in the xy -plane?

ID: e0f59119 Answer

Correct Answer: 14.66, 14.67, 44/3

Rationale

The correct answer is $\frac{44}{3}$. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the xy -plane and b is the y -intercept. Distributing the $\frac{1}{3}$ in the equation $y = \frac{1}{3}29x + 10 + 5x$ yields $y = \frac{29}{3}x + \frac{10}{3} + 5x$. Combining like terms on the right-hand side of this equation yields $y = \frac{44}{3}x + \frac{10}{3}$. This equation is in the form $y = mx + b$, where $m = \frac{44}{3}$ and $b = \frac{10}{3}$. Therefore, the slope of the graph of the given equation in the xy -plane is $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 44b2b894

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■ ■ ■

ID: 44b2b894

SAT6 M 2.20

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1,089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

ID: 44b2b894 Answer

Correct Answer: 66

Rationale

The correct answer is 66. It's given that each vertex of the rectangle lies on the circumference of the circle. Therefore, the length of the diameter of the circle is equal to the length of the diagonal of the rectangle. The diagonal of a rectangle forms a right triangle with the shortest and longest sides of the rectangle, where the shortest side and the longest side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. Let s represent the length, in units, of the shortest side of the rectangle. Since it's given that the diagonal is twice the length of the shortest side, $2s$ represents the length, in units, of the diagonal of the rectangle. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting s for a and $2s$ for c in this equation yields $s^2 + b^2 = 2s^2$, or $s^2 + b^2 = 4s^2$. Subtracting s^2 from both sides of this equation yields $b^2 = 3s^2$. Taking the positive square root of both sides of this equation yields $b = s\sqrt{3}$. Therefore, the length, in units, of the rectangle's longest side is $s\sqrt{3}$. The area of a rectangle is the product of the length of the shortest side and the length of the longest side. The lengths, in units, of the shortest and longest sides of the rectangle are represented by s and $s\sqrt{3}$, and it's given that the area of the rectangle is $1,089\sqrt{3}$ square units. It follows that $1,089\sqrt{3} = ss\sqrt{3}$, or $1,089\sqrt{3} = s^2\sqrt{3}$. Dividing both sides of this equation by $\sqrt{3}$ yields $1,089 = s^2$. Taking the positive square root of both sides of this equation yields $33 = s$. Since the length, in units, of the diagonal is represented by $2s$, it follows that the length, in units, of the diagonal is 233, or 66. Since the length of the diameter of the circle is equal to the length of the diagonal of the rectangle, the length, in units, of the diameter of the circle is 66.

Question Difficulty: Hard

Question ID e723bd67

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e723bd67

SAT6 M 2.21

$2x - y > 883$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
440	0
441	-2
442	-4

B.

x	y
440	0
442	-2
441	-4

C.

x	y
442	0
440	-2
441	-4

D.

x	y
442	0
441	-2
440	-4

ID: e723bd67 Answer

Correct Answer: D

Rationale

Choice D is correct. All the tables in the choices have the same three values of x , 440, 441, and 442, so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting 440 for x in the given inequality yields $2440 - y > 883$, or $880 - y > 883$. Subtracting 880 from both sides of this inequality yields $-y > 3$. Dividing both sides of this inequality by -1 yields $y < -3$. Therefore, when $x = 440$, the corresponding value of y must be less than -3 . Substituting 441 for x in the given inequality yields $2441 - y > 883$, or $882 - y > 883$. Subtracting 882 from both sides of this inequality yields $-y > 1$. Dividing both sides of this inequality by -1 yields $y < -1$. Therefore, when $x = 441$, the corresponding value of y must be less than -1 . Substituting 442 for x in the given inequality yields $2442 - y > 883$, or $884 - y > 883$. Subtracting 884 from both sides of this inequality yields $-y > -1$. Dividing both sides of this inequality by -1 yields $y < 1$. Therefore, when $x = 442$, the corresponding value of y must be less than 1. For the table in choice D, when $x = 440$, the corresponding value of y is -4 , which is less than -3 ; when $x = 441$, the corresponding value of y is -2 , which is less than -1 ; when $x = 442$, the corresponding value of y is 0 , which is less than 1. Therefore, the table in choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. When $x = 440$, the corresponding value of y in this table is 0 , which isn't less than -3 .

Choice B is incorrect. When $x = 440$, the corresponding value of y in this table is 0 , which isn't less than -3 .

Choice C is incorrect. When $x = 440$, the corresponding value of y in this table is -2 , which isn't less than -3 .

Question Difficulty: Medium

Question ID 9a216c0f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 9a216c0f

SAT6 M 2.22

The combined original price for a mirror and a vase is \$60. After a 25% discount to the mirror and a 45% discount to the vase are applied, the combined sale price for the two items is \$39. Which system of equations gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase?

- A. $m + v = 60$ $0.55m + 0.75v = 39$
- B. $m + v = 60$ $0.45m + 0.25v = 39$
- C. $m + v = 60$ $0.75m + 0.55v = 39$
- D. $m + v = 60$ $0.25m + 0.45v = 39$

ID: 9a216c0f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that m represents the original price, in dollars, of the mirror, and v represents the original price, in dollars, of the vase. It's also given that the combined original price for the mirror and the vase is \$ 60. This can be represented by the equation $m + v = 60$. After a 25% discount to the mirror is applied, the sale price of the mirror is 75% of its original price. This can be represented by the expression $0.75m$. After a 45% discount to the vase is applied, the sale price of the vase is 55% of its original price. This can be represented by the expression $0.55v$. It's given that the combined sale price for the two items is \$ 39. This can be represented by the equation $0.75m + 0.55v = 39$. Therefore, the system of equations consisting of the equations $m + v = 60$ and $0.75m + 0.55v = 39$ gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase.

Choice A is incorrect. The second equation in this system of equations represents a 45% discount to the mirror and a 25% discount to the vase.

Choice B is incorrect. The second equation in this system of equations represents a 55% discount to the mirror and a 75% discount to the vase.

Choice D is incorrect. The second equation in this system of equations represents a 75% discount to the mirror and a 55% discount to the vase.

Question Difficulty: Medium